

2.1.

a) $Y(0)$ is the potential outcome if not treated.

b) $Y(0)|D=1$ is the outcome if not treated for the subgroup D. D is the group of people who received treatment anyways.

$Y(0)|d=1$ is the potential outcome if not treated for the group d, that receive treatment. This group is not observable.

c) $Y(0)$ is the potential outcome if not treated. $Y(0)|D=0$ is the outcome if not treated only of the subgroup who did not get treated.

d) $Y(0)|D=1$ is the outcome for the people who were assigned to the control group but received treatment anyway. $Y(0)|D=0$ is the outcome for the people who were assigned to the control group and were not treated.

e) $E[Y(0)]$ is the average outcome of the control group. $E[Y(0)|D=1]$ is the average outcome for the control group for the group that

f) $E[Y(0)|D=1] - E[Y(0)|D=0]$ is the average potential outcome for untreated group among people if they did receive treatment minus the average outcome for untreated group for people who did not receive treatment.

If the result is zero, it means that without any treatment, the groups would have performed identically. Any difference must be due to the treatment.

2.10

a. Excludability means that outcomes are solely explained through receiving the treatment, not through the process of assignment.

If a letter is sent, subjects are aware that they are in the study. People could become more interested in politics due to the letter sent. The effect measures therefore is a combination of the letter and receiving the newspaper.

b. The non-interference assumption says that a subject's outcome must only relate to their assigned group, not another subject's assigned group.

By bringing newspapers to the cafeteria, there is a spillover. The outcome of someone in the control group now also depends on another subject's assignment.

2.12

a. The equation shows zero selection bias. Zero selection bias can be achieved when there is random assignment. In this specific case, nature decides about the two groups, which means that there is no random assignment, therefore one can not assume that there was no selection bias in this study.

b. The excludability assumption means that an outcome is solely explained through receiving the treatment. The act of separating the prisoners from the others mixes up whether the outcome relates to the change in routing or the single act of reading.

c. Non interference means that a subjects outcome should not relate to a different subjects assignment. Their potential outcome should not be affected by any other subject reading or not reading. The assumption holds in the experiment.

d. Treating everyone changes the whole environment of the prison, which creates spillover effects. This means that the outcome is no longer the isolated outcome of reading, but of a altered prison environment. The outcome for one subject doesn't depend on them being treated or not anymore, but on different things (like an altered prison environment).